

Divyesh A. Mistry, PhD

Computational Materials Scientist

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Professional Summary

Computational materials scientist specializing in molecular dynamics, multiscale mechanics, and high-performance scientific computing. Develops LAMMPS, discrete dislocation dynamics, and Python workflows that connect atomistic mechanisms to measurable deformation, strengthening, adhesion, and fracture response. Experience spans Ni-based superalloys, tungsten and W-Re alloys, polymer adhesives, constitutive modeling, defect characterization, and reproducible analysis on HPC clusters. Published first-author research and open simulation software; available for computational materials, scientific software, and mechanics roles.

Technical Skills

Simulation & Modeling: Molecular dynamics (MD), discrete dislocation dynamics (DDD), finite element analysis (FEA/FEM), multiscale modeling, constitutive modeling, fracture mechanics, plasticity, microstructure evolution.

Simulation Software: LAMMPS, OVITO, AtomsK, ABAQUS, ANSYS, COMSOL, NEPER, ParaView, MATLAB.

Programming & Scientific Python: Python (NumPy, Pandas, SciPy, Matplotlib), C++, Fortran, Bash, workflow automation, large-scale data analysis, Git/GitHub, L^AT_EX.

High-Performance Computing (HPC): Linux/Unix, MPI/OpenMP parallel computing, HPC cluster management, batch scheduling, multi-million-atom simulations.

Professional Experience

Postdoctoral Researcher

University of Seville

Aug 2026–Present

Seville, Spain

- Develop molecular-dynamics and multiscale workflows for cross-linked epoxy and polymer adhesive systems, linking network chemistry and crosslink density to thermomechanical response, glass-transition temperature, shear strength, and failure.
- Design reproducible LAMMPS protocols for curing, equilibration, cooling, and deformation; evaluate thermostat, strain-rate, and finite-size effects to improve verification and physical interpretation.
- Build Python pipelines for replica-based statistical analysis, stress-strain processing, property extraction, uncertainty assessment, and automated visualization of large simulation campaigns on HPC systems.
- Connect atomistic shear response with cohesive-zone and continuum descriptions of adhesive joints to support strength and life-prediction models across length scales.
- **Project:** *Multiscale modeling of interfacial delamination of steel/adhesive interfaces by aqueous corrosion*, funded by the U.S. Office of Naval Research.
- **Advisors:** Prof. María del Pilar Ariza Moreno and Prof. Michael Ortiz.

Postdoctoral Fellow / Visiting Researcher

Merrimack College, Department of Mechanical and Electrical Engineering

Apr 2026–Jul 2026

North Andover, MA, USA

- Performed large-scale MD simulations of temperature-dependent crack-tip deformation and the ductile-to-brittle transition in tungsten and W-Re alloys.
- Developed automated LAMMPS and Python workflows for stress-strain analysis, fracture-energy calculations, defect characterization, dislocation extraction, and trajectory processing on HPC clusters.
- Integrated atomistic fracture mechanisms with a 3D polycrystalline DDD framework to study Frank-Read source activation, dislocation multiplication, grain-boundary interactions, and plastic shielding.
- Mentored student research and contributed to manuscript development and multiscale simulation planning.

Junior Research Fellow, DRDO-Sponsored Project

Indian Institute of Technology Bombay

Jan 2025–Dec 2025

Mumbai, India (concurrent with final thesis defense)

- Investigated constitutive behavior and high-strain-rate response of shape-memory-alloy composites for aerospace and defense applications.
- Conducted Split Hopkinson Pressure Bar experiments and used experimental data to calibrate and validate numerical constitutive models.
- Supported project execution, data analysis, technical reporting, and mentoring of junior researchers.

PhD Researcher, Aerospace Engineering

Indian Institute of Technology Bombay

Jan 2019–Jun 2025

Mumbai, India

- Developed an atomistically informed multiscale framework for dislocation-precipitate-interface interactions in powder-metallurgy Ni-based superalloys.
- Executed multi-million-atom LAMMPS simulations to quantify generalized stacking-fault energies, dislocation mobility, precipitate shearing, critical resolved shear stress, and temperature-dependent mechanisms.
- Established size-dependent power laws for paired edge dislocations interacting with γ' precipitates and transferred atomistic drag and obstacle-strength parameters into DDD models.
- Developed polycrystalline DDD/finite-element models of prior particle boundaries, quantifying hardening, pinned

dislocation density, geometrically necessary dislocations, and residual-stress retention.

- Created parallel and automated Python/C++/Fortran workflows for simulation setup, defect tracking, post-processing, verification, parameter studies, and publication-quality visualization.
- Published first-author research, presented at international conferences, and collaborated with computational and experimental researchers.

Graduate Teaching Assistant

Indian Institute of Technology Bombay

Jan 2019–Dec 2023

Mumbai, India

- Supported Continuum Mechanics, Data Analysis, and Multiscale Modeling courses; developed computational exercises and guided undergraduate and postgraduate students.

Assistant Professor, Mechanical Engineering

CMR Institute of Technology

Jun 2016–Dec 2018

Bengaluru, India

- Taught Finite Element Methods, Experimental Stress Analysis, and Solid Mechanics; developed laboratory modules using ANSYS, MATLAB, and Python.

Graduate Teaching Assistant and MTech Researcher

Indian Institute of Technology Kharagpur

Jun 2014–Jul 2016

Kharagpur, India

- Developed and validated ABAQUS and ANSYS models for structural mechanics, smart composites, and vibration; supported laboratories in FEM, vibration analysis, and tribology.

Education

PhD, Aerospace Engineering, Indian Institute of Technology Bombay 2019–2025

Thesis: *Multiscale Modeling of Prior Particle Boundaries in Nickel-Based Superalloys*; CGPA: 9.36/10

Advisors: Prof. P. J. Guruprasad and Prof. Amuthan A. Ramabathiran

MTech, Mechanical Engineering (Mechanical System Design), IIT Kharagpur 2014–2016

Thesis: *A Mesh-Free Model for Static Analysis of Smart Composite Beams*; CGPA: 8.85/10

BE, Aeronautical Engineering (Aero Mechanical), Aeronautical Society of India 2013

Publications

Peer-Reviewed Journal Articles

1. **Mistry, D. A.**, Tak, T. N., Sankarasubramanian, R., and Guruprasad, P. J. “A Discrete Dislocation Dynamics Study of Prior Particle Boundary Effects in Ni-Based Superalloys.” *Journal of Engineering Materials and Technology* (2026). DOI.
2. **Mistry, D. A.** and Ramabathiran, A. A. “Size-Dependent Power Laws for Edge Dislocations in Nickel Superalloys: A Molecular Dynamics Study.” *Computational Materials Science* 259, 114122 (2025). DOI.

Manuscripts in Preparation

1. **Mistry, D. A.** and Mahata, A. “Atomistic Indicators of the Ductile-to-Brittle Transition in Polycrystalline Tungsten: Temperature and Rhenium Effects on Crack-Tip Plasticity.” arXiv preprint arXiv:2608.27589 (2026). Preprint.
2. **Mistry, D. A.**, Tak, T. N., and Guruprasad, P. J. “Polycrystalline Discrete Dislocation Dynamics of Ni-Based Superalloys: Interactions with Prior Particle Boundaries and Second Phases Using Atomistically Informed Inputs.” Manuscript in preparation.

Conference Contributions

1. EMI 2026, Boulder, USA: *Multiscale Modeling of Dislocation–Precipitate–PPB Interactions in Powder-Metallurgy Nickel Superalloys*.
2. IMPLAST 2025, IIT Roorkee, India: *Hierarchical Multiscale Modeling of Plasticity in Ni-Based Superalloys*.
3. COMPLAS 2023, Barcelona, Spain: *An Atomistically Informed DDD Study of Prior Particle Boundaries in Ni-Based Superalloys*.

Awards, Grants & Professional Memberships

- **Institution of Eminence Travel Grant**, IIT Bombay (2025) — funded attendance at TMS 2025, USA.
- **IIT Bombay International Travel Grant** (2023) — funded attendance at COMPLAS 2023, Spain.
- **GATE All India Rank: 69** (2014) — Graduate Aptitude Test in Engineering; MHRD fellowships for doctoral and master’s study.
- **Member**, The Minerals, Metals & Materials Society (TMS); **Associate Member**, Aeronautical Society of India (AeSI).
- Certifications: Using Python for Research (HarvardX); The Unix Workbench (Johns Hopkins); Material Behavior (Georgia Tech); Engineering Simulations (CornellX).

Additional Information

Research interests: computational materials science, multiscale mechanics, atomistic simulation, deformation and fracture, additive-manufacturing materials, polymer adhesives, and data-driven materials modeling. Academic references available upon request.